Unit 8: Surface Area & Volume of 3D Solids — Penguin Huddles, Camping Tents, Aquariums & Ziplines

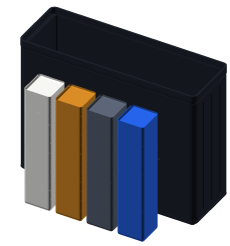
Surface Area of Rectangular & Triangular Prisms, Pyramids, Volume of Prisms, and Circles/Cones/Hypotenuse

Why doesn't an Emperor penguin chick freeze when Antarctic winds hit -40°C? When Zoe read that penguins survive by huddling together to shrink their exposed 'surface area' while keeping their warm 'volume' the same, she admitted she always mixed those two terms up in math class. Liam had the same headache in the workshop: when he sewed nylon for a camping tent or ordered glass for an open-top fish tank, he wasn't sure when to add flat walls and when to multiply length times width times height.

Marcus ran into a surprise of his own at the team pizza night. The pizzeria offered either one 16-inch pizza or two 10-inch pizzas for the same price, and almost everyone at the table assumed two 10-inch pizzas had more food simply because 10 + 10 is 20.

Out on the adventure course, they also needed to find the exact length of a diagonal zipline cable stretched from the top of a 15-meter tower to an anchor 20 meters away on the ground without climbing into the air with a tape measure.

By unfolding cardboard boxes and tents flat onto the table as 2D nets, packing wooden cubes into rectangular trays, and checking prism volumes, circle areas, and right-triangle diagonals on the SC-6, Zoe, Liam, and Marcus master every 3D measurement.



Companion Story in From Blocks to the Stack (SC-6 Elementary Guide) • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + 3x1 Storage Boxes

UNIT 8 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

C

SEC 8.1 (pp.77-78) Three-Dimensional Figures, Nets & Rectangular Prisms

Textbook Ways: Way 1: Unfold into a 6-Face 2D Net | Way 2: Opposite Face Pairs SA = 2(lw + lh + wh) | Way 3:..

Learn by Hand: The surface area of a polyhedron is the sum of the areas of all its flat faces, which can be visualized by unfolding the solid into a two-dimensional net. A rectangular prism of length l, width w, and height h has 6 rectangular faces arranged in 3 congruent opposite pairs: SA = 2lw + 2lh + 2wh.

Check Your Work on the SC-6 Calculator:

[g] [CLΣ] 8 [ENTER] 5 [f] [■] 2 [×] [Σ+] 8 [ENTER] 3 [f] [■] 2 [×] [Σ+] 5 [ENTER] 3 [f] [■] 2 [×] [Σ+] [f] [Σx]

Unfold the 6-face net and find the area of each pair on paper first. On your SC-6, clear stats with [g][CLΣ], add each face pair using [f][RECT] 2 [×] [Σ+], and press [f][Σx] to confirm 158 cm²!

SEC 8.2 (pp.79-80) Surface Areas of Triangular Prisms and Pyramids

Textbook Ways: Way 1: Add All Individual Faces on the 2D Net | Way 2: Base Area + Combined Lateral Area

Learn by Hand: To find the surface area of a triangular prism, unfold its net into 2 congruent triangular bases and 3 rectangular lateral faces, and sum their areas. To find the surface area of a pyramid, add the area of its single polygon base to the areas of its triangular lateral faces (using the slant height of each triangle).

Check Your Work on the SC-6 Calculator:

[g] [CLΣ] 1 0 [x²] [Σ+] 1 0 [ENTER] 1 2 [g] [▲] 4 [×] [Σ+] [f] [Σx]

Press [g][CLΣ] 10 [x²] [Σ+] 10 [ENTER] 12 [g][TRI] 4 [×] [Σ+] [f][Σx] to confirm 340 m²!

SEC 8.3 (pp.81-82) Volumes of Rectangular Prisms

Textbook Ways: Way 1: Base Area × Height (V = Bh) | Way 2: Multiply Length × Width × Height (Fractions or De..

Learn by Hand: While Surface Area measures the 2D flat area covering the outside of a solid (in square units), Volume measures the 3D space inside the solid—the number of unit cubes needed to fill it completely (in cubic units). For any rectangular prism, Volume equals the base area B times the height h: V = Bh = l × w × h.

Check Your Work on the SC-6 Calculator:

1 2 [ENTER] 8 [ENTER] 5 [f] [▣]

Type 12 [ENTER] 8 [ENTER] 5 [f][PRISM] to confirm 480 in³ in one step!

SEC 8.4 (pp.83-84) Circles, Cones, and the Pythagorean Theorem

Textbook Ways: Way 1: Exact Form in Terms of π & a² + b² = c² | Way 2: Decimal Approximation (π ≈ 3.14) & Scal..

Learn by Hand: In this Grade 7/8 Geometry Bridge: (1) The area of a circle with radius r is A = πr² (where r is half the diameter d). (2) In a right triangle with legs a and b and hypotenuse c, the Pythagorean Theorem states $a^2 + b^2 = c^2$, so $c = \sqrt{(a^2 + b^2)}$. (3) The volume of a cone is one-third the volume of a cylinder with the same base and height: $V = 1/3\pi r^2 h$.

Check Your Work on the SC-6 Calculator:

6 [f] [●] 1 2 [ENTER] 5 [g] [HYP]

Press 6 [f][CIRC] to verify 113.10 in², and 12 [ENTER] 5 [g][HYP] to verify 13 ft!

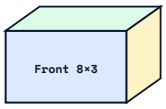
HANDS-ON MODEL: 3X1 GRIDFINITY BOX & TILES

Section 8.1 — Practice A: Three-Dimensional Figures, Nets & Rectangular Prisms

Key Vocabulary: polyhedron • face • edge • vertex • rectangular prism • cube • net • surface area (SA) Name: _____ Date: _____ Score: ____ / 100

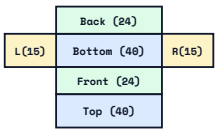
TEXTBOOK CONCEPT & HANDS-ON MODEL (3X1 GRIDFINITY BOX & TILES)

3D Prism (8×5×3 cm)



SA = 2(40+24+15) = 158

Way 1: Unfolded 6-Face 2D Net



HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (2D Net): Unfold the 3D rectangular prism into 6 flat rectangles (Top, Bottom, Front, Back, Left, Right) and add all 6 areas.

Textbook Method 2 (Opposite Face Pairs Formula): Since opposite faces are congruent, use SA = 2lw + 2lh + 2wh = 2(lw + lh + wh).

WATCH OUT: A rectangular prism has 6 faces (3 matching pairs: 2lw + 2lh + 2wh) — don't forget to double all three face pairs!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS) C

Problem: Find the surface area of a rectangular prism measuring 8 cm long, 5 cm wide, and 3 cm high.

WAY 1: Add All 6 Faces on a 2D Net

Top (40) + Bottom (40) + Front (24) + Back (24) + Left (15) + Right (15) = 80 + 48 + 30 = 158 cm².

WAY 2: Opposite Face Pairs Formula SA = 2(lw + lh + wh)

Sum the 3 distinct face areas and multiply by 2: SA = 2(8·5 + 8·3 + 5·3) = 2(40 + 24 + 15) = 2(79) = 158 cm².

WAY 3: Lateral Perimeter Strip + 2 Bases (Ph + 2B)

4 side walls form a strip of width P = 2(8+5) = 26 cm and height 3 cm (26×3 = 78 cm²), plus 2 bases (80 cm²) = 158 cm².

Answer: 158 cm² 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!) C

Worked Example SC-6 Check Keystrokes:

g [CLΣ] 8 ENTER 5 f ■ 2 × [Σ+] 8 ENTER 3 f ■ 2 × [Σ+] 5 ENTER 3 f ■ 2 × [Σ+] [f] [Σx]

Unfold the 6-face net and find the area of each pair on paper first. On your SC-6, clear stats with [g][CLΣ], add each face pair using [f][RECT] 2 [×] [Σ+], and press [f][Σx] to confirm 158 cm²!

SECTION 8.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Find the surface area of a cube with edge length s = 4 cm.

Starter Step 1: A cube has 6 identical square faces—square the edge length (4^2) to find 1 face, then multiply by 6.

1 Square Face = s² = _____

Cube SA = 6 × s² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Find the surface area of a cube with edge length s = 9 in.

Starter Step 1: A cube has 6 identical square faces—square the edge length (9^2) to find 1 face, then multiply by 6.

1 Square Face = s² = _____

Cube SA = 6 × s² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Find the surface area of a rectangular prism measuring 6 cm long, 4 cm wide, and 3 cm high.

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 Find the surface area of a rectangular box measuring 10 in. long, 5 in. wide, and 2 in. high.

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 Find the surface area of a rectangular prism with length l = 12 ft, width w = 8 ft, and height h = 5 ft.

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 Find the surface area of a rectangular prism measuring 5 m long, 4 m wide, and 2.5 m high.

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 A cube has a total surface area of 150 cm². What is the area of one square face, and what is the edge length s of the cube?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 Error Analysis: For a 5 cm by 4 cm by 3 cm box, Leo added 20 + 15 + 12 = 47 cm². What did he forget to do, and what is the correct surface area?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 Open-Top Cold-Storage Battery Cargo Pod: An open-top rectangular battery cargo pod (5 faces, no top lid) is 30 cm long, 20 cm wide, and 15 cm tall. How many square centimeters of insulation paneling are needed to line the bottom and four sides?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Gift Box Wrapping Paper: You are wrapping a gift box that is 15 inches long, 10 inches wide, and 4 inches tall. If you have a sheet of wrapping paper with an area of 550 in.², do you have enough paper to cover the box?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: 3X1 GRIDFINITY BOX & TILES

Section 8.1 — Practice B: Modeling Real Life (Three-Dimensional Figures, Nets & Rectangular Prisms)

Textbook Ways: Unfold into a 6-Face 2D Net | Opposite Face Pairs SA = 2(lw + lh + wh) | Lateral Strip Ph + 2B Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — CUBE & OPEN-TOP STORAGE BIN (2 TEXTBOOK WAYS = +20 PTS DOUBLE) C

Problem: Find the surface area of a cube with edge length 7 inches, and find the outer surface area of an open-top bin measuring 10 cm by 8 cm by 5 cm.

WAY 1: Add the Exposed Faces on the 2D Net

Cube (6 identical 7×7 squares): 6 × 49 = 294 in². Open-Top Bin (5 faces): 1 bottom (10×8 = 80) + 2 fronts (2×50 = 100) + 2 sides (2×40 = 80) = 260 cm².

Cube Formula: 6 congruent 7 × 7 squares -> SA = 6s² = 6(49) = 294 in².

WAY 2: Full 6-Face Formula Minus the Open Top

Full 10×8×5 prism SA = 2(80 + 50 + 40) = 340 cm². Subtract the missing 10×8 top lid (80 cm²): 340 - 80 = 260 cm².

Open-Top Box (5 Faces): 1 bottom (80) + 2 fronts (100) + 2 sides (80) = 260 cm² (or 340 - 80 = 260 cm²).

WATCH OUT: A rectangular prism has 6 faces (3 matching pairs: 2lw + 2lh + 2wh) — don't forget to double all three face pairs!

Answer: Cube = 294 in²; Open-Top Bin = 260 cm²

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!) C

Worked Example SC-6 Check Keystrokes:

[7] [x^2] [6] [×]

Calculate 6s² for the cube and the 5 panels of the open-top bin on paper first. Check on your SC-6:
Press 7 [x^2] 6 [×] to confirm 294 in² for the cube.

Liam used to plug numbers blindly into the 6-wall box formula SA = 2(lw + lh + wh) without thinking about the actual object—until he ordered too much glass for an open-top aquarium and got the wrong painted area when Zoe glued two 4-centimeter wooden cubes face-to-face to model huddling penguins. When they unfolded a paper model of the fish tank flat on the table, Liam saw why the formula failed him: an open-top tank only has 5 glass panels (no top lid!), and gluing two cubes together hides their 2 touching faces inside, leaving only 10 exposed outer squares instead of 12.

SECTION 8.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Cereal Box Cardboard: A cereal box is a rectangular prism measuring 20 cm long, 8 cm wide, and 30 cm tall. What is its total surface area?**

Starter Step 1: Find the area of the bottom (20×8), front (20×30), and side (8×30) rectangles, add them, and multiply by 2.

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Foam Die Cube: A giant foam die is a cube with edge length 9 inches. What is the area of one face, and what is the total surface area of all 6 faces?**

Starter Step 1: Multiply 9 in. × 9 in. to find the area of 1 square face, then multiply by 6 faces for the total surface area.

1 Square Face = s² = _____

Cube SA = 6 × s² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Open-Top Glass Aquarium: A rectangular glass fish tank is 24 inches long, 12 inches wide, and 15 inches tall, with an open top (no glass lid). What is the total area of the 5 glass panels?**

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Wrapping Paper Check: You want to wrap a game console box measuring 15 in by 10 in by 4 in using a sheet of wrapping paper with area 600 in². Find the surface area of the box. Do you have enough paper?**

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Painting a Bedroom: A bedroom is 12 feet long, 10 feet wide, and 8 feet high. You paint the 4 walls and the ceiling, but not the floor. How many square feet do you paint?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Cube Edge from Surface Area: A wooden cube has a total surface area of 150 square centimeters. What is the area of one face, and what is the edge length of the cube?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Storage Trunk: A rectangular storage trunk measures 4 feet by 2 feet by 2 feet. What is its total surface area?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Juice Box Carton: A juice box measures 6 cm by 4 cm by 10 cm. What is its total surface area?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Doubling the Edge of a Cube: Cube A has an edge length of 3 cm and Cube B has an edge length of 6 cm. Find the surface area of each cube. How many times as great is Cube B's surface area?**

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Two Cubes Glued Together: Two 4 cm by 4 cm by 4 cm cubes are glued together face-to-face to form an 8 cm by 4 cm by 4 cm rectangular prism. What is the surface area of the new prism?**

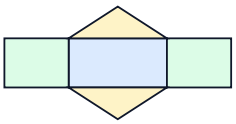
CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Section 8.2 — Practice A: Surface Areas of Triangular Prisms and Pyramids

Key Vocabulary: triangular prism • pyramid • lateral face • slant height • regular pyramid • 2D net Name: _____ Date: _____ Score: ____ / 100

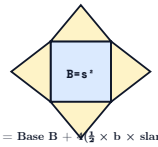
TEXTBOOK CONCEPT & HANDS-ON MODEL (COUNTING & ARRAY TILES)

Triangular Prism Net (5 Faces)



SA = 2 Tri Bases ($\frac{1}{2}$bh) + 3 Rect Sides

Square Pyramid Net (1 Base + 4 Triangles)



SA = Base B + 4($\frac{1}{2}$ × b × slant l)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Triangular Prisms: Unfold into 2 congruent triangular bases (each $\frac{1}{2}$bh) plus 3 rectangular lateral faces, and add all 5 areas.

Textbook Pyramids: Unfold into 1 polygon base B plus triangular lateral faces using the slant height l (each $\frac{1}{2}$bl): SA = B + 4($\frac{1}{2}$bl).

WATCH OUT: Use the $\frac{1}{2}$ on EVERY triangular face (A = $\frac{1}{2}$bh or $\frac{1}{2}$bl), and count all faces in the net (5 for triangular prism, 5 for square pyramid)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

C

Problem: Find the surface area of a square pyramid with a 10 m by 10 m square base and 4 congruent triangular faces each having base 10 m and slant height 12 m.

WAY 1: Add All 5 Faces on the Pyramid Net

Square base = 10 × 10 = 100 m². Each triangular face = $\frac{1}{2}$ × 10 × 12 = 60 m². Total SA = 100 + 60 + 60 + 60 + 60 = 340 m².

WAY 2: Base Area + 4 × Lateral Triangle Formula

Since all 4 lateral triangles are congruent: SA = B + 4($\frac{1}{2}$bl) = 10² + 4($\frac{1}{2}$ · 10 · 12) = 100 + 240 = 340 m².

Answer: 340 m² 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

C

Worked Example SC-6 Check Keystrokes:

[g] [CLΣ] 1 0 [x²] [Σ+] 1 0 [ENTER] 1 2 [g] [▲] 4 [×] [Σ+] [f] [Σx]

Add the square base (10²) and 4 triangles on paper first. Check on your SC-6:
Press [g][CLΣ] 10 [x²] [Σ+] 10 [ENTER] 12 [g][TRI] 4 [×] [Σ+] [f][Σx] to confirm 340 m²!

SECTION 8.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Find the surface area of a square pyramid with a 6 cm by 6 cm base and 4 triangular faces each having base 6 cm and slant height 5 cm.

Starter Step 1: Unfold the pyramid into 1 square base (6 × 6) plus 4 identical triangles (each A = $\frac{1}{2}$ × 6 × 5), then add them.

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Find the surface area of a square pyramid with an 8 in. by 8 in. base and 4 triangular faces each having base 8 in. and slant height 10 in.

Starter Step 1: Unfold the pyramid into 1 square base (8 × 8) plus 4 identical triangles (each A = $\frac{1}{2}$ × 8 × 10), then add them.

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Find the surface area of a square pyramid with a 12 ft by 12 ft base and 4 triangular faces each having base 12 ft and slant height 9 ft.

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 Four-Triangle Pyramid (Tetrahedron): A 4-sided pyramid is made of 4 identical equilateral triangles, each with a base of 6 cm and a height of 5.2 cm. What is the total surface area of all 4 faces?

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 A triangular prism has 2 right-triangle bases (b = 4 cm, h = 3 cm) and 3 rectangular faces of length 10 cm (widths 3 cm, 4 cm, and 5 cm). What is its surface area?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 A triangular prism tent has 2 triangular ends (base 8 ft, height 3 ft) and 3 rectangular panels of length 7 ft (widths 5 ft, 5 ft, and 8 ft). What is its total surface area?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 A regular tetrahedron has 4 congruent triangular faces each with base 8 cm and slant height 6.93 cm. What is its surface area?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 Compare the nets of a triangular prism and a square pyramid: how many triangular faces and how many rectangular/square faces does each solid have?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 A-Frame Camping Tent: An A-frame tent is a triangular prism with 2 triangular doors (base 6 ft, height 4 ft), a 6 ft by 8 ft floor, and 2 sloped roof panels (5 ft by 8 ft each). How many square feet of nylon fabric are in the tent?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Glass Pyramid Skylight: A rooftop skylight is shaped like a square pyramid with an open bottom (no floor)—only 4 congruent triangular glass panes, each with base 9 ft and slant height 6 ft. What is the total area of the glass panes?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: COUNTING & ARRAY TILES

Section 8.2 — Practice B: Modeling Real Life (Surface Areas of Triangular Prisms and Pyramids)

Textbook Ways: Add All Individual Faces on the 2D Net | Base Area + Combined Lateral Area Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SKATEBOARD RAMP (TRIANGULAR PRISM) (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

C

Problem: A skate ramp is a triangular prism with 2 right-triangle ends (base 6 ft, height 8 ft, ramp 10 ft) and a width of 10 ft. Find its total surface area.

WAY 1: Add All 5 Faces on the Unfolded 2D Net

2 Triangles = $2 \times (\frac{1}{2} \cdot 6 \cdot 8) = 48$ ft². 3 Rectangles = $(6\times10) + (8\times10) + (10\times10) = 60 + 80 + 100 = 240$ ft². Sum: $48 + 240 = 288$ ft².

5-Face Net Sum: 2 triangles ($24 + 24 = 48$ ft²) + 3 rectangles ($100 + 100 + 120 = 320$ ft²) = 368 ft².

WAY 2: Combine the 3 Rectangles into One Lateral Strip

Perimeter of triangle base = $6 + 8 + 10 = 24$ ft. Lateral rectangle = $24 \times 10 = 240$ ft². Add 2 triangle bases (48 ft²) -> 288 ft².

Lateral Strip Check: $2B + P{\cdot}L = 48 + (10 + 10 + 12) \times 10 = 48 + 320 = 368$ ft².

WATCH OUT: Use the ½ on EVERY triangular face (A = ½bh or ½bl), and count all faces in the net (5 for triangular prism, 5 for square pyramid)!

Answer: 288 square feet

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

C

Worked Example SC-6 Check Keystrokes:

6 ENTER 8 g ▲ 2 × 6 ENTER 8 + 1 0 + 1 0 f ■ +

Add the 2 triangular ends and 3 rectangular sides on paper first. Check on your SC-6:
Press 6 [ENTER] 8 [g][TRI] 2 [×] (48 ft²) plus 6 [ENTER] 8 [+] 10 [+] 10 [f][RECT] (240 ft²) [+] to confirm 288 ft².

When Liam tried to calculate the waterproof fabric needed for a triangular-prism camping tent and the shingles for a square-pyramid gazebo roof, he felt overwhelmed by all the 3D edges and slant heights. He was afraid he would count the wrong number of sides or forget to divide a triangle by 2. Then Marcus suggested mentally 'unzipping' the tent and flattening every wall onto the floor as a 2D net: 1 rectangular floor, 2 rectangular roof slopes, and 2 triangular end doors (5 faces total). Counting and labeling each flat piece in the net before doing any arithmetic kept him from missing a single wall.

SECTION 8.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Camping Tent: A tent shaped like a triangular prism has 2 triangular doors (each with base 6 ft and height 4 ft), two 5 ft by 8 ft roof panels, and a 6 ft by 8 ft rectangular floor. What is the total surface area of the tent?**

Starter Step 1: Find the area of the 2 triangular doors (each ½ × 6 × 4), the 2 rectangular roof panels (5 × 8), and the floor (6 × 8).

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Glass Pyramid Model: A scale model of a square pyramid has a 12 cm by 12 cm square base and 4 congruent triangular faces, each with a slant height of 10 cm. What is its total surface area?**

Starter Step 1: Find the area of the 12 cm by 12 cm square base plus the 4 congruent triangular faces (each A = ½ × 12 × 10).

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Gazebo Roof Shingles: A gazebo roof is a square pyramid with base edge 8 feet and slant height 6 feet. Find the lateral surface area (the 4 triangular roof faces, not including the open bottom).**

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Triangular Chocolate Box: A triangular prism box has 2 equilateral triangle ends (each with area 6.9 cm²) and 3 identical rectangular sides (each 4 cm by 20 cm). What is the total surface area of the box?**

Base Area(s) B = _____

+ Lateral Faces = _____ -> SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Wooden Wedge: A right-triangular prism wedge has 2 triangular bases with legs 8 in and 6 in and hypotenuse 10 in. The prism is 5 inches wide. What is its total surface area?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Four-Sided Game Die: A pyramid-shaped 4-sided die (called a tetrahedron) has 4 identical triangular faces, each with a base of 3 cm and a height of 2.6 cm. What is its total surface area?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Attic Roof & Gables: An attic prism has 2 triangular gables (each base 16 ft, height 6 ft) and 2 rectangular roof slopes (each 10 ft by 30 ft). What is the combined area of the 2 gables and 2 roof slopes?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Pyramid Paperweight: A square pyramid has a 6 in by 6 in base and a total surface area of 96 in². What is the area of one of its four triangular faces, and what is its slant height?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Triangular Prism Gift Box: A prism has 2 triangular bases (each with base 12 cm and height 8 cm) and 3 rectangular faces of widths 10 cm, 10 cm, and 12 cm, each 15 cm long. What is its total surface area?**

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Comparing Two Pyramids: Square Pyramid A has a 10 cm by 10 cm base and slant height 8 cm. Square Pyramid B has an 8 cm by 8 cm base and slant height 10 cm. Which pyramid has the greater total surface area?**

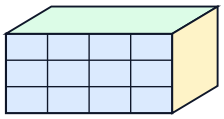
CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Section 8.3 — Practice A: Volumes of Rectangular Prisms

Key Vocabulary: volume (V) • cubic units • base area (B = lw) • V = Bh = lwh • fractional edge length Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (COUNTING & FRACTION STUDIO)

Unit-Cube Layer Packing



Base B = 12×8 = 96 | × 5 layers = 480 in³

3 Ways to Find Prism Volume:

Way 1 (Layers): V = Base Area B × h

Way 2 (Formula): V = l × w × h

Missing Height: h = V ÷ (l × w)

(Wait: 1/2-in cubes -> 8 fit per 1 in³!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Volume measures the number of unit cubes that fill a 3D solid. Multiply Base Area B = lw by height h: V = Bh = lwh.

Textbook Fractional Edge Lengths: Pack unit-fraction cubes OR multiply fractional dimensions as improper fractions or decimals.

WATCH OUT: Volume is measured in CUBIC units (in.³, cm³). Eight $\frac{1}{2}$-inch cubes ($2 \times 2 \times 2 = 8$) fit inside 1 cubic inch!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

C

Problem: Find the volume of a crate measuring 12 in. by 8 in. by 5 in., and an aquarium measuring 3 1/2 ft by 1 1/2 ft by 2 ft.

WAY 1: Layer Stacking V = Bh (With Unit Cubes)

Crate: Base B = 12 × 8 = 96 in²; multiply by 5 layers -> 480 in³. Tank ($3\frac{1}{2} \times 1\frac{1}{2} \times 2$): Base B = (7/2)(3/2) = 21/4; × 2 = 10 $\frac{1}{2}$ ft³.

WAY 2: Multiply Three Dimensions (Improper Fractions or Decimals)

Crate: 12 × (8 × 5) = 12 × 40 = 480 in³. Tank in decimals: 3.5 × 1.5 × 2 = 3.5 × 3 = 10.5 ft³ (10 $\frac{1}{2}$ ft³).

Answer: Crate = 480 in.³; Aquarium = 10 1/2 ft³

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

C

Worked Example SC-6 Check Keystrokes:

1 2 ENTER 8 ENTER 5 f [cube]

Multiply length × width × height on paper first. Then check on your SC-6 using the 3-number Prism Volume key: Type 12 [ENTER] 8 [ENTER] 5 f[PRISM] to confirm 480 in³ in one step!

SECTION 8.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Cereal Storage Container: Find the volume of a rectangular plastic container that is 18 cm long, 6 cm wide, and 28 cm tall.

Starter Step 1: Multiply length × width (18 × 6) to find the Base Area B, then multiply by the height (28 cm).

Base Area B = l × w = _____

Volume V = B × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Find the volume of a cube with edge length s = 5 inches.

Starter Step 1: For a cube with edge s = 5 in., multiply length × width × height: V = 5 × 5 × 5.

Cube Edge s = _____

Volume V = s³ = s×s×s = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Camp Lap Pool: A rectangular camp swimming pool is 25 meters long, 10 meters wide, and filled with water 2 meters deep. How many cubic meters of water does it hold?

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 A rectangular prism has a base area of B = 32 m² and a height of h = 7 m. What is its volume?

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 Find the volume of a fish tank that is 3 1/2 feet long, 1 1/2 feet wide, and 2 feet tall.

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 Find the volume of a wooden block measuring 4 1/2 inches long, 3 inches wide, and 2 inches high.

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 A rectangular storage box has a volume of 270 in.³, a length of 9 in., and a width of 5 in. What is its height h?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 How many 1/2-inch cubes are needed to completely fill a 1-inch cube (measuring 1 in. by 1 in. by 1 in.)?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 Garden Planter Depth: A raised cedar planter has a base measuring 9 inches by 5 inches and holds 360 cubic inches of potting soil. What is the depth h of the soil?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Packing Half-Inch Cubes: A wooden box has a volume of 27 cubic inches. How many 1/2-inch unit cubes can be packed inside the box?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Section 8.3 — Practice B: Modeling Real Life (Volumes of Rectangular Prisms)

Textbook Ways: Base Area × Height (V = Bh) | Multiply Length × Width × Height (Fractions or Decimals) Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — FINDING MISSING DEPTH FROM VOLUME (2 TEXTBOOK WAYS = +20 PTS DOUBLE) C

Problem: A cedar planter box is 9 inches long and 5 inches wide, and holds 360 cubic inches of soil. How deep is the soil in the planter box?

WAY 1: Find Base Area B First, Then Divide V ÷ B

Base area B = 9 × 5 = 45 in². Substitute into V = Bh: 360 = 45h -> h = 360 ÷ 45 = 8 inches deep.

Missing Height Formula: V = Bh -> 1,440 = (15 × 12)h -> 1,440 = 180h -> h = 1,440 ÷ 180 = 8 inches.

WAY 2: Divide Volume by Length, Then by Width

Divide 360 ÷ 9 in = 40 in², then divide 40 ÷ 5 in = 8 inches deep.

Layer Check: 180 in² base × 8 in height = 1,440 in³.

WATCH OUT: Volume is measured in CUBIC units (in.³, cm³). Eight ½-inch cubes (2 × 2 × 2 = 8) fit inside 1 cubic inch!

Answer: 8 inches deep

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!) C

Worked Example SC-6 Check Keystrokes:

3 6 0 ENTER 4 5 ÷

Find the base area B = 9 × 5 = 45 in² and divide 360 ÷ 45 on paper first. Check on your SC-6:
Press 360 [ENTER] 45 [÷] to confirm a depth of 8 inches.

Zoe used to confuse Surface Area (square units, cm²) with Volume (cubic units, cm³) until she watched a crow drop pebbles into a rectangular water tube with an 8 cm by 5 cm base. Wrapping paper or glass goes on the flat outside walls (square centimeters), while water and pebbles fill up the 3D space inside (cubic centimeters). Even better, she saw that every rectangular prism fills up like stackable layers: one 1-centimeter-deep layer of water in the tube holds 8 × 5 = 40 cm³, so when the pebbles push 120 cm³ of water upward, dividing 120 ÷ 40 shows the water must rise exactly 3 centimeters!

SECTION 8.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Swimming Lap Pool: A rectangular pool is 25 meters long, 10 meters wide, and 2 meters deep. What is the volume of the pool?**

Starter Step 1: Multiply length × width × depth (25 m × 10 m × 2 m) to find the volume in cubic meters.

Base Area B = l × w = _____

Volume V = B × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Raised Garden Bed: A cedar garden bed is 8 feet long, 4 feet wide, and 1 1/2 feet deep. How many cubic feet of soil are needed to fill it?**

Starter Step 1: Write 1 1/2 as 1.5 (or 3/2), then multiply length × width × depth (8 × 4 × 1.5).

Base Area B = l × w = _____

Volume V = B × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **3D Printer Build Volume: A 3D printer has a cubic build volume measuring 25 cm on each edge. What is its build volume in cubic centimeters?**

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Bento Lunchbox: A rectangular bento box measures 6 1/2 inches long, 4 inches wide, and 2 1/2 inches tall. What is its volume?**

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Half-Inch Cube Packing: How many 1/2-inch cubes fit inside a 1-cubic-inch space, and how many 1/2-inch cubes are needed to completely fill a 3 in by 2 in by 2 in box?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Crow Water-Displacement Tank (Aesop's Fable): A crow's rectangular water tube has an 8 cm by 5 cm base. Dropping pebbles into the water displaces 120 cm³ of volume. By how many centimeters does the water level rise?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Cargo Container: A rectangular cargo box is 5.5 feet long, 4 feet wide, and 3.5 feet tall. What is its volume?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Concrete Patio Slab: A rectangular patio is 12 feet long, 9 feet wide, and 1/3 foot (4 inches) thick. How many cubic feet of concrete are needed?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Acrylic Puzzle Box: A kea parrot's acrylic puzzle box has a rectangular base of 9 inches by 5 inches and a volume of 360 cubic inches. What is the height of the puzzle box?**

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Aquarium Gallons: A fish tank measures 24 inches long, 12 inches wide, and 16 inches tall. Find its volume in cubic inches, and then find how many gallons of water it holds (1 gallon = 231 in³).**

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 8.4 — Practice A: Circles, Cones, and the Pythagorean Theorem

Key Vocabulary: circle • radius (r) • diameter (d = 2r) • circumference (C = 2πr) • area (A = πr²) • cone (V = 1/3πr²h) • hypotenuse

Name: _____ Date: _____ Score: _____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Circle: A = πr²



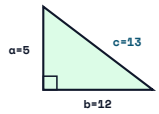
r = d ÷ 2 | 36π ~ 113.10

Cone: V = 1/3πr²h



Cone = 1/3 of Cylinder!

Right Triangle: a²+b²=c²



c = √(5²+12²) = √169 = 13

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Circle & Cone Formulas: Circle Area A = πr² (if given diameter d, divide by 2 first: r = d/2). Cone Volume V = 1/3Bh = 1/3πr²h.

Textbook Pythagorean Theorem: In a right triangle with legs a and b and hypotenuse c, a² + b² = c², so c = √(a² + b²).

WATCH OUT: If given diameter d, divide by 2 FIRST to get radius r = d ÷ 2 before squaring r²! And don't forget the 1/3 for cones!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

C

Problem: Find the area of a circular pizza with radius r = 6 in. (round to 2 decimal places), and find the diagonal length of a ramp with legs 12 ft and 5 ft.

WAY 1: Formula Evaluation (Exact π & Square Root)

Circle (r = 6): A = π(6²) = 36π ~ 113.10 in². Right triangle (legs 12, 5): c² = 12² + 5² = 144 + 25 = 169 -> c = √169 = 13 ft.

WAY 2: Decimal Estimate (π ~ 3.14) & Pythagorean Triples

Circle: 36 × 3.14 = 113.04 in² (or 36 × 3.1416 = 113.10 in²). Triangle: recognize the 5-12-13 Pythagorean triple -> c = 13 ft.

Answer: Pizza = 113.10 in.²; Ramp diagonal = 13 ft

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

C

Worked Example SC-6 Check Keystrokes:

6 f ● 1 2 ENTER 5 g

Cut the 12-inch diameter in half (r = 6) and calculate πr² and √(12² + 5²) on paper first. Check on your SC-6: Press 6 [f][CIRC] to verify 113.10 in², and 12 [ENTER] 5 [g][HYP] to verify 13 ft!

SECTION 8.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Find the area of a circle with radius r = 5 cm (round to the nearest hundredth).

Starter Step 1: By PEMDAS, square the radius FIRST (r² = 5 × 5), and then multiply by π.

r = _____ | r² = _____

Area A = π × r² ~ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Garden Sundial: A circular garden sundial has a diameter of 30 cm across the center. Find its area to the nearest hundredth.

Starter Step 1: Divide the 30 cm diameter by 2 to find the radius r FIRST, then square the radius and multiply by π.

r = _____ | r² = _____

Area A = π × r² ~ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Find the hypotenuse c of a right triangle with legs a = 3 cm and b = 4 cm.

a² + b² = ___ + ___ = ___

c = √(a² + b²) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 Treehouse Zip-Line Shortcut: A backyard treehouse platform is 6 meters high, and the anchor post is 8 meters away across flat ground. How long is the straight diagonal zip-line cable connecting them?

r = _____ | r² = _____

Area A = π × r² ~ _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 A skateboard ramp has a horizontal base of 12 feet and a vertical back height of 5 feet. How long is the sloped riding surface?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 A circular landing pad has a diameter of d = 16 meters. What is its radius r, and what is its area rounded to the nearest hundredth?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 Conical Sand Filtration Pile: A conical pile of water-filtration sand at the treatment plant has a base radius of r = 3 meters and a height of h = 2 meters. What is the volume of the sand pile, rounded to the nearest hundredth?

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 Find the volume of a cone with radius r = 6 inches and height h = 5 inches, rounded to the nearest hundredth.

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 Cylinder vs. Cone Volume: A cylinder and a cone both have radius r = 3 cm and height h = 10 cm. Find the volume of the cylinder and the volume of the cone, rounded to the nearest hundredth.

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Widescreen Monitor Diagonal: Computer monitors are advertised by their diagonal screen length. If a monitor screen is 32 inches wide and 24 inches tall, what is its diagonal measurement?

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 8.4 — Practice B: Modeling Real Life (Circles, Cones, and the Pythagorean Theorem)

Textbook Ways: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | Decimal Approximation (π ≈ 3.14) & Scaled Triples Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — VOLUME OF A CONE (2 TEXTBOOK WAYS = +20 PTS DOUBLE) C

Problem: A waffle cone has a radius of 3 centimeters and a height of 10 centimeters. Find the volume of the cone rounded to the nearest hundredth.

WAY 1: Cone Formula $V = 1/3\pi r^2 h$
Base area $B = \pi(3^2) = 9\pi$ cm². Multiply by 1/3 × height: $V = 1/3(9\pi)(10) = 3\pi \times 10 = 30\pi \approx 94.25$ cm³.

Cone Volume: $V = 1/3\pi(3^2)(10) = 1/3(90\pi) = 30\pi \approx 94.25$ cm³.

WAY 2: Find Cylinder Volume First, Then Divide by 3
A cylinder with r = 3 and h = 10 has $V = \pi(3^2)(10) = 90\pi \approx 282.74$ cm³. Divide by 3: 282.74 ÷ 3 ≈ 94.25 cm³.

Cylinder Comparison: Cylinder = $90\pi \approx 282.74$ cm³ (3 cones = 1 cylinder!).

WATCH OUT: If given diameter d, divide by 2 FIRST to get radius r = d ÷ 2 before squaring r^2! And don't forget the 1/3 for cones!

Answer: 94.25 cm³ (exact: 30π cm³)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!) C

Worked Example SC-6 Check Keystrokes:
3 ENTER 1 0 g ▲

Calculate $V = 1/3\pi r^2 h$ on paper first. Then check on your SC-6 using the Cone Volume key:
Type radius 3 [ENTER] height 10 [g][CONE] to confirm 94.25 cm³!

At the team pizza party, Marcus and his friends were sure that two 10-inch diameter pizzas had more pizza than one 16-inch diameter pizza—until they opened the boxes and saw how huge the 16-inch pizza really was. Working it out on a napkin, Marcus caught the two traps that fool almost everyone with circles: first, a 16-inch pizza has a diameter of 16 inches, so you MUST cut it in half to get the radius (r = 8 inches, not 16!); second, because the area formula squares the radius ($A = \pi r^2$), one 8-inch radius gives $64\pi \approx 201$ in², easily beating two 5-inch radius pizzas ($2 \times 25\pi \approx 157$ in²).

SECTION 8.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Trampoline Mat: A circular trampoline has a radius of 7 feet. What is the area of the jumping mat, rounded to the nearest hundredth?**
Starter Step 1: Square the 7-foot radius FIRST (7 × 7), and then multiply by π to find the circular mat's area.

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Ferris Wheel Circle: A Ferris wheel has a diameter of 40 meters. What is its radius, and what is the area enclosed by the wheel (to the nearest hundredth)?**
Starter Step 1: Divide the 40-meter diameter by 2 to find the radius r, then square r and multiply by π.

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Bicycle Tire Distance: A bicycle wheel has a radius of 14 inches from the center hub to the tire tread. How many inches forward does the bike roll in one full turn of the wheel (its circumference, to the nearest hundredth)?**

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Zipline Cable: A zipline tower is 15 meters tall and is anchored to the ground 20 meters away horizontally. How long is the diagonal zipline cable?**

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Monitor Diagonal: A rectangular computer monitor screen is 16 inches wide and 12 inches tall. What is its diagonal screen size?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Snow Cone Cup: A paper snow cone cup is shaped like a cone with radius 4 cm and height 9 cm. What is its volume (to the nearest hundredth)?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Highway Safety Cone: A traffic cone has a base diameter of 12 inches and a height of 15 inches. What is its volume (to the nearest hundredth)?**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Pizza Size Comparison: Does one 16-inch diameter pizza or two 10-inch diameter pizzas give you more total pizza area? Justify your answer by finding the areas.**

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Basketball Court Semicircle: The top of a basketball free-throw lane is a semicircle with a radius of 6 feet. What is the area of the semicircle (to the nearest hundredth)?**

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Ladder Against a Wall: The bottom of a ladder is placed 8 feet from the base of a wall so that the top of the ladder reaches 15 feet up the wall. How long is the ladder?**

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 8: Surface Area & Volume of 3D Solids — Practice A & B Complete Key

Big Ideas Math Grade 6 — Chapter 8 • Hands-On Models: Counting & Array Tiles (Counting & Array Tiles) + 3x1 Storage Boxes

Target Misconception: Students drop one of the 6 faces when finding rectangular prism surface area, tangle prism and pyramid formulas, and confuse 2D Surface Area (cardboard outside) with 3D Volume (cubes inside).

Hands-On Block Remediation: Have students touch the 3 pairs of faces on their physical box while logging each pair into [Σ+] (n=1, 2, 3), use [f] [PRISM] ([f][9]) to multiply 3 stack floors (l, w, h) at once for volume, and unlock [f] [CIRC], [g] [CONE], and [g] [HYP].



Double-Points Mastery Scoring: +10 points = Textbook Method 1 (1st Way); +20 points = Verified with Textbook Method 2 (2nd Way)!

SEC 8.1 (pp.77-78): THREE-DIMENSIONAL FIGURES, NETS & RECTANGULAR PRISMS

Methods: W1: Unfold into a 6-Face 2D Net | W2: Opposite Face Pairs SA = 2(lw + lh + wh) | W3: Lateral Strip Ph + 2B

PRACTICE A (p.77): KEY IDEAS (#1-#10)

ExA: 158 cm²

A1: 96 cm²
Setup: $4^2 \times 6$
SC-6: 4 [x²] 6 [×]

A2: 486 in.²
Setup: $9^2 \times 6$
SC-6: 9 [x²] 6 [×]

A3: 108 cm²
Setup: 2(24)+2(18)+2(12)
SC-6: Mental / Table Check

A4: 160 in.²
Setup: 2(50)+2(20)+2(10)
SC-6: Mental / Table Check

A5: 392 ft²
Setup: 2(96)+2(60)+2(40)
SC-6: Mental / Table Check

A6: 85 m²
Setup: 2(20)+2(12.5)+2(10)
SC-6: Mental / Table Check

A7: One face = 25 cm²; s = 5 cm
Setup: $\sqrt{(150 \div 6)}$
SC-6: 150 [ENTER] 6 [÷] [f][√x]

A8: 94 cm² (He only summed 3 faces instead of doubling for all 6..
Setup: 47 × 2
SC-6: 47 [ENTER] 2 [×]

A9: 2,100 cm²
Setup: 600 + 2(450) + 2(300)
SC-6: Mental / Table Check

A10: 500 in.² needed; Yes, with 50 in.² to spare
Setup: 2(150)+2(60)+2(40)
SC-6: Mental / Table Check

PRACTICE B (p.78): REAL LIFE (#1-#10)

ExB: Cube = 294 in²; Open-Top Bin = 260 cm²

B1: 2(160 + 600 + 240) = 2,000 cm²
Setup: (((20 × 8) + (20 × 30)) + (8 × 30)) × 2
SC-6: 20 [ENTER] 8 [×] 20 [ENTER] 30 [×] [+] 8 [ENTER] 30 [×] [+] 2..

B2: 1 face = 81 in²; Total SA = 486 in²
Setup: $9^2 \times 6$
SC-6: 9 [x²] 6 [×]

B3: 288 + 720 + 360 = 1,368 in²
Setup: ((24 × 12) + ((24 × 15) × 2)) + ((12 × 15) × 2)
SC-6: 24 [ENTER] 12 [×] 24 [ENTER] 15 [×] 2 [×] [+] 12 [ENTER] 15 [×..

B4: SA = 500 in²; yes (100 in² to spare)
Setup: (((15 × 10) + (15 × 4)) + (10 × 4)) × 2
SC-6: 15 [ENTER] 10 [×] 15 [ENTER] 4 [×] [+] 10 [ENTER] 4 [×] [+] 2..

B5: 120 + 192 + 160 = 472 ft²
Setup: ((12 × 10) + ((12 × 8) × 2)) + ((10 × 8) × 2)
SC-6: 12 [ENTER] 10 [×] 12 [ENTER] 8 [×] 2 [×] [+] 10 [ENTER] 8 [×]..

B6: 1 face = 25 cm²; edge = 5 cm
Setup: $\sqrt{(150 \div 6)}$
SC-6: 150 [ENTER] 6 [÷] [f][√x]

B7: 32 + 8 = 40 ft²
Setup: ((4 × 2) × 4) + ((2 × 2) × 2)
SC-6: 4 [ENTER] 2 [×] 4 [×] 2 [ENTER] 2 [×] 2 [×] [+]

B8: 2(24 + 60 + 40) = 248 cm²
Setup: (((6 × 4) + (6 × 10)) + (4 × 10)) × 2
SC-6: 6 [ENTER] 4 [×] 6 [ENTER] 10 [×] [+] 4 [ENTER] 10 [×] [+] 2 [×]

B9: 54 cm² vs. 216 cm² (4 times as great)
Setup: (6^2 × 6) ÷ 54
SC-6: 6 [x²] 6 [×] 54 [÷]

B10: 160 cm²
Setup: $4^2 \times 10$
SC-6: 4 [x²] 10 [×]

SEC 8.2 (pp.79-80): SURFACE AREAS OF TRIANGULAR PRISMS AND PYRAMIDS

Methods: W1: Add All Individual Faces on the 2D Net | W2: Base Area + Combined Lateral Area

PRACTICE A (p.79): KEY IDEAS (#1-#10)

ExA: 340 m²

A1: 96 cm²
Setup: $6^2 + (\frac{1}{2}(6\times5) \times 4)$
SC-6: 6 [x²] 6 [ENTER] 5 [g][TRI] 4 [×] [+]

A2: 224 in.²
Setup: $8^2 + (\frac{1}{2}(8\times10) \times 4)$
SC-6: 8 [x²] 8 [ENTER] 10 [g][TRI] 4 [×] [+]

A3: 360 ft²
Setup: $12^2 + (\frac{1}{2}(12\times9) \times 4)$
SC-6: 12 [x²] 12 [ENTER] 9 [g][TRI] 4 [×] [+]

A4: 62.4 cm²
Setup: $\frac{1}{2}(6\times5.2) \times 4$
SC-6: 6 [ENTER] 5.2 [g][TRI] 4 [×]

A5: 132 cm²
Setup: $\frac{1}{2}(4\times3) \times 2$
SC-6: 4 [ENTER] 3 [g][TRI] 2 [×] (12) + 12×10 (120)

A6: 150 ft²
Setup: $\frac{1}{2}(8\times3) \times 2$
SC-6: 8 [ENTER] 3 [g][TRI] 2 [×] (24) + 18×7 (126)

A7: 110.88 cm²
Setup: $\frac{1}{2}(8\times6.93) \times 4$
SC-6: 8 [ENTER] 6.93 [g][TRI] 4 [×]

A8: Triangular prism: 2 triangles + 3 rectangles; Square pyramid..
Setup: Count net faces
SC-6: Mental / Table Check

A9: 152 ft²
Setup: 24 + 48 + 80
SC-6: Mental / Table Check

A10: 108 ft²
Setup: $\frac{1}{2}(9\times6) \times 4$
SC-6: 9 [ENTER] 6 [g][TRI] 4 [×]

PRACTICE B (p.80): REAL LIFE (#1-#10)

ExB: 288 square feet

B1: 24 + 80 + 48 = 152 ft²
Setup: $(\frac{1}{2}(6\times4) \times 2) + ((5\times8) \times 2)) + (6\times8)$
SC-6: 6 [ENTER] 4 [g][TRI] 2 [×] 5 [ENTER] 8 [f][RECT] 2 [×] [+] 6 [..

B2: 144 + 240 = 384 cm²
Setup: $12^2 + (\frac{1}{2}(12\times10) \times 4)$
SC-6: 12 [x²] 12 [ENTER] 10 [g][TRI] 4 [×] [+]

B3: 96 ft²
Setup: $\frac{1}{2}(8\times6) \times 4$
SC-6: 8 [ENTER] 6 [g][TRI] 4 [×]

B4: 13.8 + 240 = 253.8 cm²
Setup: (6.9 × 2) + ((4×20) × 3)
SC-6: 6.9 [ENTER] 2 [×] 4 [ENTER] 20 [f][RECT] 3 [×] [+]

B5: 48 + 120 = 168 in²
Setup: $(\frac{1}{2}(8\times6) \times 2) + (((8 + 6) + 10) \times 5)$
SC-6: 8 [ENTER] 6 [g][TRI] 2 [×] 8 [ENTER] 6 [+] 10 [+] 5 [×] [+]

B6: 15.6 cm²
Setup: $\frac{1}{2}(3\times2.6) \times 4$
SC-6: 3 [ENTER] 2.6 [g][TRI] 4 [×]

B7: 96 + 600 = 696 ft²
Setup: $(\frac{1}{2}(16\times6) \times 2) + ((10\times30) \times 2)$
SC-6: 16 [ENTER] 6 [g][TRI] 2 [×] 10 [ENTER] 30 [f][RECT] 2 [×] [+]

B8: 15 in² per face; slant height = 5 in
Setup: (96 - 36) ÷ 4
SC-6: 96 [ENTER] 36 [-] 4 [÷]

B9: 96 + 480 = 576 cm²
Setup: $(\frac{1}{2}(12\times8) \times 2) + (((10 + 10) + 12) \times 15)$
SC-6: 12 [ENTER] 8 [g][TRI] 2 [×] 10 [ENTER] 10 [+] 12 [+] 15 [×] [+]

B10: Pyramid A (260 cm² > 224 cm²)
Setup: $8^2 + (\frac{1}{2}(8\times10) \times 4)$
SC-6: 10 [x²] 10 [ENTER] 8 [g][TRI] 4 [×] [+] 8 [x²] 8 [ENTER] 10 [g..

SEC 8.3 (pp.81-82): VOLUMES OF RECTANGULAR PRISMS

Methods: W1: Base Area × Height (V = Bh) | W2: Multiply Length × Width × Height (Fractions or Decimals)

PRACTICE A (p.81): KEY IDEAS (#1-#10)

ExA: Crate = 480 in.³; Aquarium = 10 1/2 ft³

A1: 3,024 cm³
Setup: Prism(18×6×28)
SC-6: 18 [ENTER] 6 [ENTER] 28 [f][PRISM]

A2: 125 in.³
Setup: 5^3
SC-6: 5 [x³]

A3: 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

A4: 224 m³
Setup: 32 × 7
SC-6: 32 [ENTER] 7 [×]

A5: 10 1/2 ft³ (10.5)
Setup: Prism(3.5×1.5×2)
SC-6: 3.5 [ENTER] 1.5 [ENTER] 2 [f][PRISM]

A6: 27 in.³
Setup: Prism(4.5×3×2)
SC-6: 4.5 [ENTER] 3 [ENTER] 2 [f][PRISM]

A7: 6 in.
Setup: 270 ÷ (9 × 5)
SC-6: 270 [ENTER] 9 [ENTER] 5 [×] [÷]

A8: 8 half-inch cubes
Setup: 2^3
SC-6: 2 [x³]

A9: 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

A10: 216 half-inch cubes
Setup: 27 × 2^3
SC-6: 27 [ENTER] 2 [x³] [×]

PRACTICE B (p.82): REAL LIFE (#1-#10)

ExB: 8 inches deep

B1: 500 m³
Setup: Prism(25×10×2)
SC-6: 25 [ENTER] 10 [ENTER] 2 [f][PRISM]

B2: 48 ft³
Setup: Prism(8×4×1.5)
SC-6: 8 [ENTER] 4 [ENTER] 1.5 [f][PRISM]

B3: 15,625 cm³
Setup: 25^3
SC-6: 25 [x³]

B4: 65 in³
Setup: Prism(6.5×4×2.5)
SC-6: 6.5 [ENTER] 4 [ENTER] 2.5 [f][PRISM]

B5: 8 per in³; 96 half-inch cubes
Setup: 12 × 8
SC-6: 12 [ENTER] 8 [×]

B6: 3 cm rise
Setup: 120 ÷ 40
SC-6: 120 [ENTER] 40 [÷]

B7: 77 ft³
Setup: Prism(5.5×4×3.5)
SC-6: 5.5 [ENTER] 4 [ENTER] 3.5 [f][PRISM]

B8: 36 ft³
Setup: (12 × 9) ÷ 3
SC-6: 12 [ENTER] 9 [×] 3 [÷]

B9: 8 inches
Setup: 360 ÷ 45
SC-6: 360 [ENTER] 45 [÷]

B10: 4,608 in³ ~ 19.95 (about 20) gallons
Setup: Prism(24×12×16) ÷ 231
SC-6: 24 [ENTER] 12 [ENTER] 16 [f][PRISM] 231 [÷]

SEC 8.4 (pp.83-84): CIRCLES, CONES, AND THE PYTHAGOREAN THEOREM

Methods: W1: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | W2: Decimal Approximation (π ~ 3.14) & Scaled Triples

PRACTICE A (p.83): KEY IDEAS (#1-#10)

ExA: Pizza = 113.10 in.²; Ramp diagonal = 13 ft

A1: 78.54 cm²
Setup: $\pi \times 5^2$
SC-6: 5 [f][CIRC]

A2: 706.86 cm²
Setup: $\pi \times 15^2$
SC-6: 15 [f][CIRC]

A3: 5 cm
Setup: $\sqrt{(3^2 + 4^2)}$
SC-6: 3 [ENTER] 4 [g][HYP]

A4: 10 meters
Setup: $\sqrt{(6^2 + 8^2)}$
SC-6: 6 [ENTER] 8 [g][HYP]

A5: 13 ft
Setup: $\sqrt{(12^2 + 5^2)}$
SC-6: 12 [ENTER] 5 [g][HYP]

A6: r = 8 m; Area = 201.06 m²
Setup: $\pi \times (16 \div 2)^2$
SC-6: 16 [ENTER] 2 [÷] [f][CIRC]

A7: 18.85 m²
Setup: $1/3\pi(3^2)(2)$
SC-6: 3 [ENTER] 2 [g][CONE]

A8: 188.50 in.³
Setup: $1/3\pi(6^2)(5)$
SC-6: 6 [ENTER] 5 [g][CONE]

A9: Cylinder = 282.74 cm³; Cone = 94.25 cm³
Setup: $1/3\pi(3^2)(10)$
SC-6: 3 [ENTER] 10 [g][CONE]

A10: 40 inches
Setup: $\sqrt{(32^2 + 24^2)}$
SC-6: 32 [ENTER] 24 [g][HYP]

PRACTICE B (p.84): REAL LIFE (#1-#10)

ExB: 94.25 cm² (exact: 30π cm²)

B1: 153.94 ft²
Setup: $\pi \times 7^2$
SC-6: 7 [f][CIRC]

B2: r = 20 m; Area ~ 1,256.64 m²
Setup: $\pi \times 20^2$
SC-6: 20 [f][CIRC]

B3: 87.96 inches
Setup: (2 × π) × 14
SC-6: 2 [ENTER] [π] [×] 14 [×]

B4: 25 meters
Setup: $\sqrt{(15^2 + 20^2)}$
SC-6: 15 [ENTER] 20 [g][HYP]

B5: 20 inches
Setup: $\sqrt{(16^2 + 12^2)}$
SC-6: 16 [ENTER] 12 [g][HYP]

B6: 150.80 cm² (48π)
Setup: $1/3\pi(4^2)(9)$
SC-6: 4 [ENTER] 9 [g][CONE]

B7: 565.49 in² (180π)
Setup: $1/3\pi(6^2)(15)$
SC-6: 6 [ENTER] 15 [g][CONE]

B8: One 16-in pizza (201.06 in² > 157.08 in²)
Setup: $\pi \times 5^2 \times 2$
SC-6: 8 [f][CIRC] 5 [f][CIRC] 2 [×]

B9: 56.55 ft³
Setup: $\pi \times 6^2 \div 2$
SC-6: 6 [f][CIRC] 2 [÷]

B10: 17 feet
Setup: $\sqrt{(8^2 + 15^2)}$
SC-6: 8 [ENTER] 15 [g][HYP]